IEOR 4404 Simulation **Fall 2025**

Homework 7

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Due Date: *Nov 12, 2025*

Problem 1 Let $U \sim Unif(0,1)$. By running 1000 simulation runs, give point estimates of $Cov(U, e^U)$ and $Corr(U, \sqrt{1-U^2})$.

Problem 2 A Markov chain $\{X_n, n \geq 0\}$ with states $0, 1, 2$, has the transition probability matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

Write the pseudo code for simulating the Markov chain up to time N (where N can be potentially a random time decided by a certain criterion; examples are in parts (e) and (f) below), by assuming each of the following:

(a) The computer has the capability to generate discrete random variables with a finite number of values, as long as we specify these values, say x_i, and probabilities, say p_i.

(b) The capability to generate $Unif(0,1)$.

By simulating the Markov chain 100 times (i.e., 100 trajectories) using any of the above pseudo codes (i.e., you may use any available built-in routine for generating discrete random variables, or use $Unif(0,1)$ generator), give point estimates of:

(c) $P(X_{10} = 1 | X_0 = 0)$

(d) $E[X_{10} | X_0 = 0]$

(e) $P(T \leq 10 | X_0 = 0)$, where $T = \min\{n : X_n = 1\}$ is the first time to hit state 1.

(f) $E[T | X_0 = 0]$, where $T = \min\{n : X_n = 1\}$ is the first time to hit state 1.

Finally, simulate the Markov chain starting from state 0, for 10000 steps. Regarding the first 100 as the burn-in steps, give point estimates of:

(g) Long-run proportions of visits to states 0, 1 and 2.

(h) Long-run average cost, where the cost is 1 for state 0, 3 for state 1, and 2 for state 2.

Problem 3 Each individual in a population of size N is, in each period, either active or inactive. If an individual is active in a period then, independent of all else, that individual will be active in the next period with probability α. Similarly, if an individual is inactive in a period then, independent of all else, that individual will be inactive in the next period with probability β.

(a) Let X_n denote the number of individuals that are active in period n. Argue that $X_n, n \geq 0$ is a Markov chain.

(b) Continuing from part (a), suppose $N = 100$, $\alpha = 0.2$ and $\beta = 0.4$. Initially there are 10 active individuals. Simulate the Markov chain $X_n, n \geq 0$ for 10000 time steps, with a burn-in period of 1000 steps, to estimate the long-run average number of active individuals. In simulating the Markov chain, you can either use any available built-in routine for generating discrete random variables, or the $Unif(0,1)$ generator.

(c) Let Y_n and Z_n denote the numbers of individuals that are active in period n that were initially active and inactive, respectively. That is, $X_n = Y_n + Z_n$ and $Y_0 = 10$ and $Z_0 = 0$. Argue that the tuple $(Y_n, Z_n), n \geq 0$ is a Markov chain.

(d) Continuing from part (c), suppose $N = 100$, $\alpha = 0.2$ and $\beta = 0.4$. Simulate the Markov chain $(Y_n, Z_n), n \geq 0$ for 10000 time steps, with a burn-in period of 1000 steps, to estimate the long-run probability that more active individuals come from the initially active group than the initially inactive group, (i.e., $\mathbb{P}(Y_n > Z_n)$). In simulating the Markov chain, you can either use any available built-in routine for generating discrete random variables, or the $Unif(0, 1)$ generator.